SPE-230145-MS

Structured Framework for Selecting Inflow Control Technologies: Bridging Subsurface Behavior and Surface Constraints

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

As reservoir complexity increases and well architectures evolve, the selection of appropriate inflow control technologies, ranging from passive inflow control devices (ICDs) to autonomous inflow control devices (AICDs) and interval control valves (ICVs), has become a multidimensional engineering challenge. Decisions must balance subsurface uncertainty, completion design limitations, surface infrastructure constraints, and long-term operational flexibility.

A six-step framework for selecting inflow control technologies in multizone completions is presented. It begins by defining project-specific objectives and constraints, followed by identifying subsurface and operational uncertainties that could compromise inflow performance. Relevant screening aspects—from technical to operational, economic, and infrastructure-based—are defined and prioritized to reflect project drivers. Viable inflow control technologies are then matched to these aspects and project objectives, enabling early elimination of unfit designs. Short-listed options are validated through modeling, analog benchmarking, or field data. The final step confirms execution readiness by aligning on implementation, monitoring plans, and life cycle adaptability.

This methodology enables consistent, transparent evaluations across technology types, including ICDs, AICDs (e.g., cyclonic, mechanical), adaptive systems (ICDs and AICDs with mechanical sleeves), ICVs (hydraulic, electrohydraulic, and electric), and hybrid configurations, while accounting for operational risks such as plugging, erosion, and intervention limitations. It enhances decision quality by promoting cross-disciplinary alignment, identifying trade-offs early, and reducing the risk of mismatched designs. Applicable across diverse reservoir conditions and field maturity levels, the framework supports scalable decision-making for both greenfield and brownfield applications.

Introduction

Inflow control technologies have evolved significantly from their origins as tools for mitigating the heel-toe effect in horizontal wells. They are now central to controlling unwanted fluid production, improving reservoir sweep, and enabling zonal selectivity in complex well architectures, unconsolidated formations, and deepwater or subsea environments.

As technology options expand, operators face increasingly complex decisions involving trade-offs between adaptability, reliability, cost, and integration with completion and surface systems. Selecting the right solution requires coordination across reservoir, production, completions, and facilities disciplines.

This paper introduces a structured, six-step framework for inflow control technology selection. The methodology integrates project-specific objectives, key uncertainties, and constraints from the reservoir through the wellbore to the surface infrastructure into a cohesive decision model that supports early multidisciplinary collaboration. It is designed to support both strategic planning and field-level implementation across a range of reservoir conditions and development strategies.

Field examples increasingly show that technology selection must move beyond isolated design choices to a structured, system-aware process that accounts for subsurface complexity, completion feasibility, and surface execution constraints through multidisciplinary evaluation. In a South American heavy oil field with strong aquifer support, cyclonic AICDs were deployed to reduce early water breakthrough, with zonal tracer monitoring confirming 57% lower water cut and increased oil recovery (Acencios et al., 2023). For a deepwater carbonate oil rim reservoir, electric ICVs were evaluated through cloud-based modeling under high geological uncertainty and surface water-handling constraints; simulation predicted a 74% reduction in water cut (Gurses et al., 2025). In a mature UAE asset, hybrid completions using ICDs with sliding sleeves enabled reconfigurable control across more than two dozen wells, demonstrating scalability under varying operational constraints (Dani, 2024). And in a giant greenfield development, a six-zone electric completion with real-time flow metering empowered choke optimization and production balancing on demand, integrating inflow and surface decision-making in near real time (Golenkin et al., 2020).

By applying this structured process, inflow control design becomes a strategic, collaborative effort, ensuring that selected technologies address both subsurface complexity and surface constraints with a consistent, transparent, and scalable methodology.

Background

Consequence of Evolving Well Architecture

Over the past three decades, the shift from vertical to horizontal and multilateral wells (Fig. 1) has transformed inflow performance management from a standalone design task into a multidisciplinary engineering challenge. Early vertical wells offered limited reservoir contact and frequently underperformed in geologically heterogeneous or thin-layered formations, where permeability contrasts and variable fluid contact were poorly controlled (Doan and Farouq Ali, 1995).

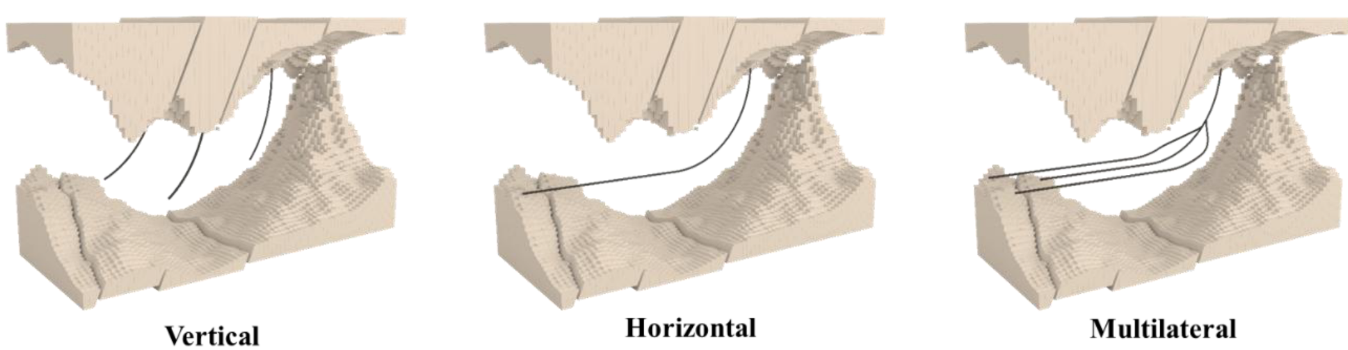


Figure 1—Evolution of well architecture, resulting in growing inflow control complexity.

The transition to horizontal wells increased reservoir contact and improved productivity, particularly under bottomhole pressure constraints or conditions conducive to coning. It also enhanced areal reservoir characterization by enabling broader data acquisition during drilling. However, uneven drawdown and friction effects in heterogeneous formations can still lead to premature water or gas breakthrough (Gurses and Vasper, 2013).

Multilateral wells expanded reservoir access and improved recovery in complex reservoirs while reducing surface footprint. These completions require careful integration of stimulation access and inflow control to manage uncertainties across branches (Salamy S. et al., 2008). Challenges become more acute in deepwater and subsea environments, where interventions are costly and surface capacity for gas handling, water disposal, and solids management is limited.

As a result, inflow management now requires integrating reservoir characterization with completion design, operational strategy, and surface system constraints to ensure performance is optimized across the full production system. The goal has shifted from merely distributing drawdown to actively managing multiphase flow, zonal conformance, and production system alignment throughout the well life.

This complexity has driven the development of a broad portfolio of inflow control technologies—passive, autonomous, adaptive, and active—to address specific combinations of reservoir heterogeneity, operational risk, and infrastructure constraints. Selecting the appropriate technology demands a holistic perspective that links subsurface uncertainty, completion design limitations, and infrastructure compatibility.

Inflow Control Technologies

Each approach to inflow control (Fig. 2) provides different capabilities with respect to drawdown distribution management, fluid selectivity, and adaptability over the life of the well. Functional principles, infrastructure needs, and intervention requirements also vary widely, making the technology selection highly context dependent.

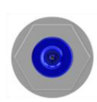
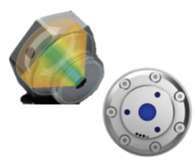
	Passive	Autonomous	Adaptive	Active
Technology	ICD (nozzle, etc.)	AICD (cyclonic, mechanical, etc.)	Sleeve + ICD/AICD combinations	ICV (hydraulic, electrohydraulic, electric)
Sample images				
Operating principle	Fixed resistance (single position)	Flow self-adjusts based on fluid type	Zonal shutoff and flow control	Surface-controlled choke with shutoff capability and multiple discrete positions or continuous displacement
Purpose	Balance inflow, delay breakthrough of unwanted fluids	Reduce unwanted fluids such as water and gas	Add intervention capability to passive and autonomous devices	Actively manage zonal contribution
Control	No adjustability after installation	Self-regulated, no surface input	Manual intervention required (via shifting tools)	Adjustable remotely from surface

Figure 2—Increasing levels of control, complexity, and operational flexibility with the progression from passive to active inflow control systems.

Passive systems such as inflow control devices (ICDs) introduce a fixed pressure drop along the wellbore to promote uniform inflow and delay early gas and water breakthrough. Their mechanical simplicity and robustness suit well-characterized, moderately heterogeneous reservoirs. However, inability to adapt post-installation limits effectiveness in dynamic or uncertain conditions.

Autonomous systems, primarily autonomous inflow control devices (AICDs), regulate flow in response to fluid properties such as density or viscosity. This paper focuses on two representative types. Cyclonic AICDs use vortex-induced centrifugal forces to divert low-viscosity fluids (e.g., gas or water) into higher-resistance paths—without moving parts. Mechanical AICDs, such as piloted floating-disk devices, incorporate internal movable elements that respond to phase changes. Both types of AICDs offer selective flow resistance without surface control, but they depend on predictable fluid contrasts and may be sensitive to mobility shifts or compositional variation along the wellbore.

Adaptive systems combine ICDs or AICDs with mechanical sleeves that allow post-installation reconfiguration via intervention. These completions offer flexibility where reservoir behavior is likely to change and periodic well interventions are feasible. However, their reliance on mechanical shifting tools introduces operational complexity and intervention risk.

Active systems, such as interval control valves (ICVs), enable real-time, surface-controlled adjustment of zonal flow. Available in hydraulic, electrohydraulic, and electric versions, ICVs support dynamic optimization throughout well life, particularly valuable in subsea or offshore platform wells where interventions are costly. Advantages include immediate zonal shutoff, adaptive drawdown management, and integration with downhole sensor systems for real-time diagnostics and active zonal control, while trade-offs include increased capital expenditure (capex), infrastructure requirements, and system complexity.

Without a structured evaluation process, technology selection often defaults to simplified assumptions, past practices, or siloed discipline input, leading to mismatches between reservoir demands and installed capabilities. Such mismatches can reduce recovery, accelerate water or gas breakthrough, or trigger costly redesigns.

A structured, cross-disciplinary framework ensures viable options are assessed consistently against project-specific goals and operational realities, aligning disciplines at an early stage to reduce nonproductive engineering time caused by late rework or misaligned assumptions. By embedding decision quality principles across the workflow, inflow control technology selection becomes a collaborative process that delivers fit-for-purpose completions and execution readiness.

Structured Framework for Inflow Control Technology Selection

As well architectures evolve and inflow control technologies diversify, selecting the optimal completion design has become increasingly complex. Traditional decision-making approaches, often heuristic, reactive, or discipline-specific, may overlook critical interdependencies between reservoir dynamics, geologic uncertainty, completion feasibility, and surface system constraints, resulting in suboptimal or non-resilient choices.

Prior research has provided valuable foundations. For example, Al-Khelaiwi et al. (2010) outlined key criteria for choosing between ICDs and ICVs based on reservoir complexity, operational flexibility, cost, and reliability, framing device selection within a performance-versus-risk perspective. Temizel et al. (2019) reviewed the evolution of smart and intelligent oilfield technologies, emphasizing the role of data integration, real-time optimization, and digital infrastructure in improving production efficiency.

Building on these foundations and experience accumulated over time, this paper presents a structured, six-step framework (Fig. 3) that integrates reservoir characterization, completion architecture, operational constraints, and project-specific objectives into a unified selection process. The methodology applies decision quality principles to promote early cross-disciplinary collaboration, transparent screening, and execution-ready inflow control solutions that are adaptable across both greenfield planning and brownfield redevelopment.

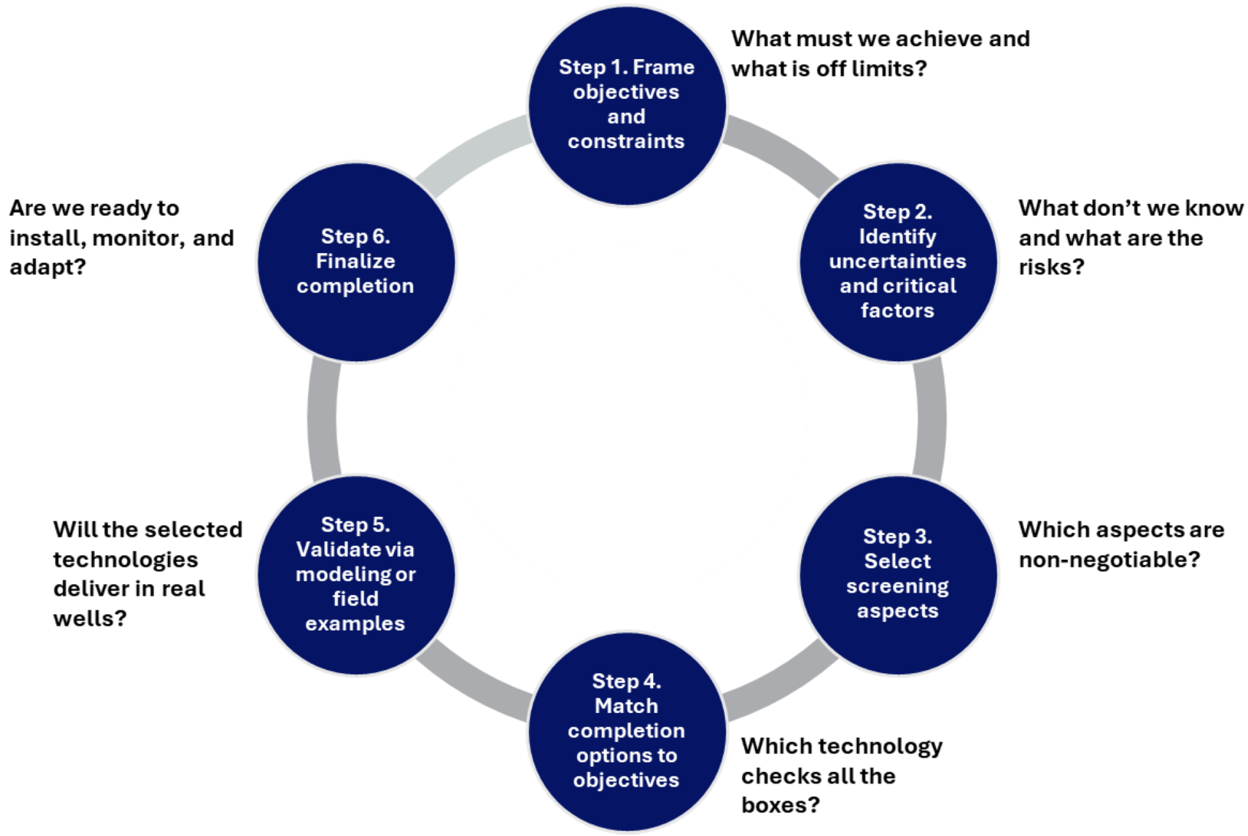


Figure 3—Structured six-step framework for inflow control technology selection.

Step 1: Frame Objectives and Constraints

What must we achieve and what is off limits?

This step defines the decision frame by aligning stakeholders—decision-makers, technical leads, and execution teams—on project goals and the boundaries within which design decisions must operate. It ensures early multidisciplinary engagement and sets a clear foundation for the subsequent screening, evaluation, and selection steps.

Project or company objectives (Fig. 4)—such as early production, managing unwanted fluids, maintaining well control, zonal selectivity, and alignment with sustainability or infrastructure goals—are defined by broader field development strategies. For example, a "Max. Recovery" strategy prioritizes zonal selectivity and monitoring; a "Low Capex" strategy favors simpler, robust solutions with performance trade-offs; and a "Fast-Track" strategy focuses on rapid deployment, potentially avoiding complex or long-lead technologies.

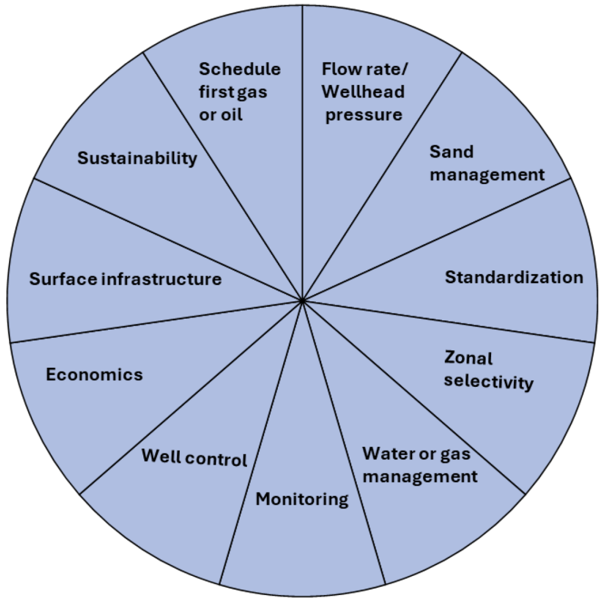


Figure 4—Key drivers and inflow control planning objectives.

Sample field development strategies with their objectives and rationales are shown in Fig. 5.

Strategy A Max. Recovery	Strategy B Low Capex	Strategy C Fast Track
• **Objective:** Maximize reservoir sweep and recovery • **Rationale:** Invest upfront for full zonal control and monitoring flexibility	• **Objective:** Minimize upfront cost • **Rationale:** Use simple, robust solutions for lower-complexity reservoirs	• **Objective:** Expedite early production • **Rationale:** Prioritize fast deployment with minimal surface impact
Hybrid Strategy A Flexible Development	**Hybrid Strategy B Cost-Conscious Recovery**	**Hybrid Strategy C Brownfield Optimization**
• **Objective:** Start simple, adapt later • **Rationale:** Enable upgrades if reservoir challenges emerge	• **Objective:** Balance cost with control • **Rationale:** Use basic technology in low-risk zones and advanced control where needed	• **Objective:** Retrofit with selectivity and minimal topside impact • **Rationale:** Deploy autonomous solutions, which reduce surface constraints while adding zone flexibility in critical areas

Figure 5—Sample field development strategies and their implications for inflow control design.

Different strategies carry distinct priorities and tolerances, influencing how inflow control systems are designed and evaluated. For instance, hybrid strategies may balance multiple objectives; for example, deploying simpler control systems during early production and upgrading later as subsurface understanding improves. However, assuming simple and robust wells (Strategy B) as the default can be misleading. An alternative is to drill fewer but higher-performing wells with more complex completions, potentially reducing overall well count and Capex, and shifting the trade-off from simplicity toward productivity per well. In both approaches, screening and ranking reservoir monitoring technologies become important to facilitate dynamic reservoir characterization. These insights can guide the placement and control logic of future wells, particularly in phased developments or infill campaigns.

Constraints, on the other hand, define the operating envelope and must be recognized and incorporated into all design decisions for viable completion designs. They include:

- **reservoir constraints,** such as heterogeneity, petrophysics, and geological uncertainty, proximity to gas and water contacts, and mobility contrasts
- **wellbore constraints** including tubing size, number of zones, intervention, and stimulation access
- **surface constraints** such as gas-handling capacity, topside space, water disposal facilities, and intervention access.

Jointly defining objectives and constraints at the beginning reduces misalignment, focuses screening, and enables early identification of risks and trade-offs, which are carried forward into Step 2.

Step 2: Identify Uncertainties and Critical Factors

What don't we know and what are the risks?. This step maps the uncertainty landscape, identifying the technical and operational risks that could compromise inflow control performance during the life of the well. While Step 1 defined project objectives and boundaries shaped by company priorities and project-specific development strategies, Step 2 focuses on how unknowns and variable conditions could limit feasibility, performance, or reliability of candidate technologies.

These risks translate into critical design factors—non-negotiable conditions that directly influence whether a particular inflow control technology is viable under the specific field conditions. For example, limited control line availability may restrict the use of hydraulic ICVs in more than two zones and a no-flaring policy may require early-phase gas control capability. Capturing these factors at an early stage ensures that nonviable options are eliminated in Step 3 and enables early mitigation planning.

All the categories listed in Table 1 influence inflow control feasibility and system selection. Each group contributes to choosing screening criteria in Step 3 by identifying risks that must be addressed during evaluation.

Table 1—Categories of key uncertainties and critical design factors.

Category	Examples of Key Uncertainties and Critical Design Factors
Reservoir uncertainties	Heterogeneity (e.g., permeability contrasts, compartmentalization, natural fractures or fissuring), breakthrough timing, mobility contrasts (water vs. oil, gas vs. oil)
Completion constraints	Zonal isolation limits, ICD and ICV reliability, flow selectivity, subsea infrastructure and control
Operational factors	Injection balance, surface handling limitations, plateau management, flow assurance challenges
Sand production risk	Risk of erosion or plugging from uncontrolled inflow, formation consolidation or breakdown, particle size distribution
Hardware and design constraints	Subsea space limitations, zone count restrictions, plugging risks
Regulatory considerations and sustainability	No flaring, water/gas reinjection, low-carbon requirements

This multidisciplinary risk identification step ensures that screening focuses not just on the field development scenario but also on the specific uncertainties and constraints that could undermine inflow control success.

Step 3: Select Screening Aspects

Which aspects are non-negotiable?

Once key uncertainties and critical design factors have been identified in Step 2, the next step is to define the screening aspects that will guide comparative evaluation of inflow control technologies. This step bridges risk awareness with practical decision-making by translating project-specific drivers into clearly defined evaluation criteria.

Rather than relying on siloed, discipline-specific checklists or assigning default weight to every possible consideration, the framework emphasizes a multidisciplinary and context-specific approach. Reservoir, completions, production, operations, and facilities engineers, together with project stakeholders, jointly define which aspects truly matter for the selected field development strategy. This ensures that both subsurface behavior and system-level execution constraints are embedded into the screening logic.

Each aspect of screening is assigned a role based on how it affects inflow control technology selection.

- The role is **performance-based (PB)** when the screening aspect evaluates the technology's ability to perform a specific technical function (e.g., gas control, cleanup). These screenings require deeper technical evaluation and comparison of system capabilities.
- The role is **constraint/elimination-based (C/EB)** when the screening aspect evaluates the technology's ability to address non-negotiable physical, operational, or economic constraints (e.g., intervention requirement, tubing size, installation time). These screenings are used to filter out technologies that are infeasible for the project conditions and objectives.

Table 2 lists a core set of 27 screening aspects grouped into five domains.

For each development strategy, only the aspects aligned with its objectives and tolerances are prioritized for active screening. For example, in offshore or deepwater settings, inflow control technologies must be evaluated for compatibility with both the reservoir and the broader field development system. Key infrastructure aspects include subsea interfacing, availability of control lines and umbilical penetrations, electric power supply, and chemical injection routing. Life cycle considerations, such as hydrate mitigation, compression timing, and monoethylene glycol (MEG) handling, must also be factored in, especially when water starts to encroach. Technology readiness level (TRL), installation time, and space allocation are other important feasibility checks.

A unified prioritization scheme (e.g., high, medium, low) is applied across all screening aspects to indicate their criticality for the selected field development strategy:

- **High:** critical to success, directly influences feasibility or performance
- **Medium:** influential but not decisive, impact can be mitigated or managed
- **Low**: limited effect on outcomes or consistently manageable across technology options.

Table 2—Screening aspects by technical domain.

Reservoir engineering	1. Uncertainty in reservoir description 2. More flexible development 3. Number of controllable zones 4. Formation permeability 5. Well type (horizontal, multilateral)
Completions engineering	6. Tubing inner diameter (ID) 7. Value of information (monitoring) 8. Zonal isolation (shutoff) 9. Equipment reliability
Production engineering	10. Control of gas 11. Control of water 12. Production or injection rate variation 13. Max. production or injection rate 14. Well cleanup efficiency 15. Acidizing and scale treatment
Economics & operations	16. Equipment cost 17. Installation complexity 18. Running-in-hole risk 19. Reservoir isolation during upper completion (UC) 20. Intervention requirement 21. Installation time 22. Plugging risk 23. Logistics constraints
Infrastructure: Surface & subsurface	24. Subsea interfacing 25. Subsea capacity constraints 26. Flow assurance risks 27. Corrosion and erosion risks

Only the most relevant aspects, those ranked High or Medium, are retained for active screening. This focuses technical evaluations on what truly drives differentiation and eliminates unfit options at an early stage. Because the importance of a screening aspect varies with reservoir complexity, infrastructure constraints, and project goals, a structured analysis of the development context is essential to support consistent, strategy-aligned prioritization.

Table 3 illustrates a comparative analysis of inflow control technologies for a selection of PB and C/EB screening aspects.

Table 3—Comparative analysis of inflow control technologies for selected aspects of screening.

Aspect	Screening Role	ICDs	AICDs (cyclonic/mechanical)	Sleeve + ICDs/AICDs	ICVs (Hydraulic/Electro-hydraulic/Electric)
1. Uncertainty in reservoir description	PB	Fixed geometry; rely on predrill models for placement effectiveness.	Add fluid discrimination; perform best with stable contrasts and known contact behavior. If reservoir fluid properties evolve over time, testing the technology performance could be required.	Offer post-installation flexibility; sleeves can be adjusted based on updated reservoir understanding.	Support real-time zonal adjustment; particularly useful in dynamic or uncertain settings.
3. Number of controllable zones	PB or C/EB	Easily scalable with packers; suitable for wells with a large number of zones.	Same as ICDs; offer high scalability in long laterals.	Moderate scalability; zone count can be increased but may introduce operational complexity.	PB: Hydraulic systems support a limited number of zones; electrohydraulic and electric ICVs enable higher zonal granularity with fewer lines. C/EB: If number of zones is 3+, hydraulic ICVs may not be feasible.
6. Tubing ID	C/EB	Preserves full tubing ID; well-suited for high flow rates and intervention access.	Same as ICDs; completion does not intrude on flow conduit.	Minor reduction in ID because of sleeves; tool access and flow capacity should be evaluated in planning.	Impact on tubing ID varies with system design; for 3+ zones, inner-string installations may reduce ID. Electric ICVs offer more compact alternatives where maintaining ID is critical.
7. Value of information (monitoring)	C/EB	No sensors by default; indirect methods (e.g., DTS, tracers) can provide inflow diagnostics but add cost and potential intervention risk.	Similar to ICDs; monitoring can be enabled through DTS or chemical tracers.	Support integration of memory gauges or tracers for zonal surveillance, particularly useful post-installation.	Combine monitoring and control; integrated sensors enable real-time decisions under evolving conditions.
20. Intervention requirement	C/EB	Fully passive system; ideal for limited-intervention environments.	Same as ICDs; no post-installation operation needed.	Reconfiguration requires mechanical shifting; may not be feasible in subsea or intervention-limited settings.	Support remote control; eliminate most intervention needs after deployment.
24. Subsea interfacing	C/EB	No control lines or power required; minimal impact on subsea hardware.	Same as ICDs; fully passive design aligns with constrained umbilical capacity.	Passive setup; requires packer and sleeve planning but no power or fluid lines.	Hydraulic systems require early planning for control lines; electric ICVs reduce subsea complexity through compact cabling. If limited number of lines are available, number of zones could be affected by the type of ICVs selected.

Aligning screening priorities at an early stage ensures that strategy-driven evaluations balance reservoir behavior, operational constraints, and project-specific goals.

Step 4: Match Completion Options to Objectives

Which inflow control technology checks all the boxes?

Building on the screening aspects prioritized in Step 3, this step evaluates inflow control technologies for the selected field development strategy to identify viable, high-value completion options. The aim is to apply a structured, transparent, and multidisciplinary evaluation process to filter configurations that best align with project goals, reservoir characteristics, operational constraints, and infrastructure limitations.

Where feasible, the evaluation begins with a broad set of options, including passive, autonomous, adaptive, active, and hybrid inflow control systems, unless prior constraints or field development decisions have already narrowed the viable technology space. This inclusive approach ensures that performance trade-offs and constraint sensitivities are objectively captured.

Candidate technologies are then assessed against the prioritized screening aspects using a traffic-light rating scheme to visualize feasibility and alignment.

- **Green = good:** Strong alignment with project requirements; no major concerns
- **Yellow = maybe:** Conditional fit; performance depends on mitigations, tuning, or further validation
- **Red = bad:** Incompatible with one or more critical constraints or objectives; not viable for the current context

This qualitative scoring enables rapid visualization of strengths and weaknesses across completion options, facilitating early elimination of unfit configurations and focused technical comparison of feasible candidates. It also improves cross-discipline communication by condensing complex screening logic into a format that clearly conveys which technologies merit deeper modeling or field validation.

For a sample subsea development, Table 4 shows how the inflow control technologies available to the operator score versus the screening aspects examined in Table 3.

Table 4—Scores for inflow control technologies across prioritized screening aspects.

Aspect	Screening Role	Priority	Key Consideration	ICDs	Cyclonic AICDs	Sleeve + ICDs	2 zones, Hydraulic ICVs	Comments
1. Uncertainty in reservoir description	PB	High	Sensitivity to subsurface uncertainty	🔴	🟢	🟡	🟢	ICDs are not adaptable post-installation. AICDs autonomously adapt. Sleeves offer some flexibility if intervention is possible. ICVs allow dynamic control.
3. Number of controllable zones	PB or C/EB	High	How many zones can be managed?	🟢	🟢	🟢	🟡	ICDs and AICDs are easily scalable with packers. Sleeves allow scalability with operational complexity. 2-zone hydraulic ICVs enable a limited number of zones, could be applicable in a certain section of reservoir, need further analysis to justify the value.
6. Tubing ID	C/EB	Medium	ID preserved for high-rate production	🟢	🟢	🟢	🟢	All these configurations preserve tubing ID (full bore), posing no constraints on expected high-rate production.
7. Value of information (monitoring)	C/EB	High	Compatibility with downhole sensors and monitoring	🔴	🔴	🔴	🟢	ICVs integrate with downhole gauges and real-time monitoring tools, enabling immediate zonal diagnostics and control. Others do not include built-in surveillance but can be complemented by external monitoring technologies such as tracers or distributed fiber-optic sensing when feasible.
20. Intervention requirement	C/EB	High	Elimination of intervention	🟢	🟢	🔴	🟢	Sleeved completions are eliminated if intervention is not acceptable.
24. Subsea interfacing	C/EB	High	Feasibility of integrating with subsea architecture	🟢	🟢	🟡	🟢	ICDs and AICDs are simple to deploy subsea. Sleeves introduce operational complexity; 2 zones with ICVs acceptable with control line limits.

The output of Step 4 is a short list of technically and operationally viable inflow control systems, each developed into a conceptual-level tubular configuration (Fig. 6) showing the number of zones and device placement, which will be carried forward into Step 5 for detailed modeling or analog-based validation.

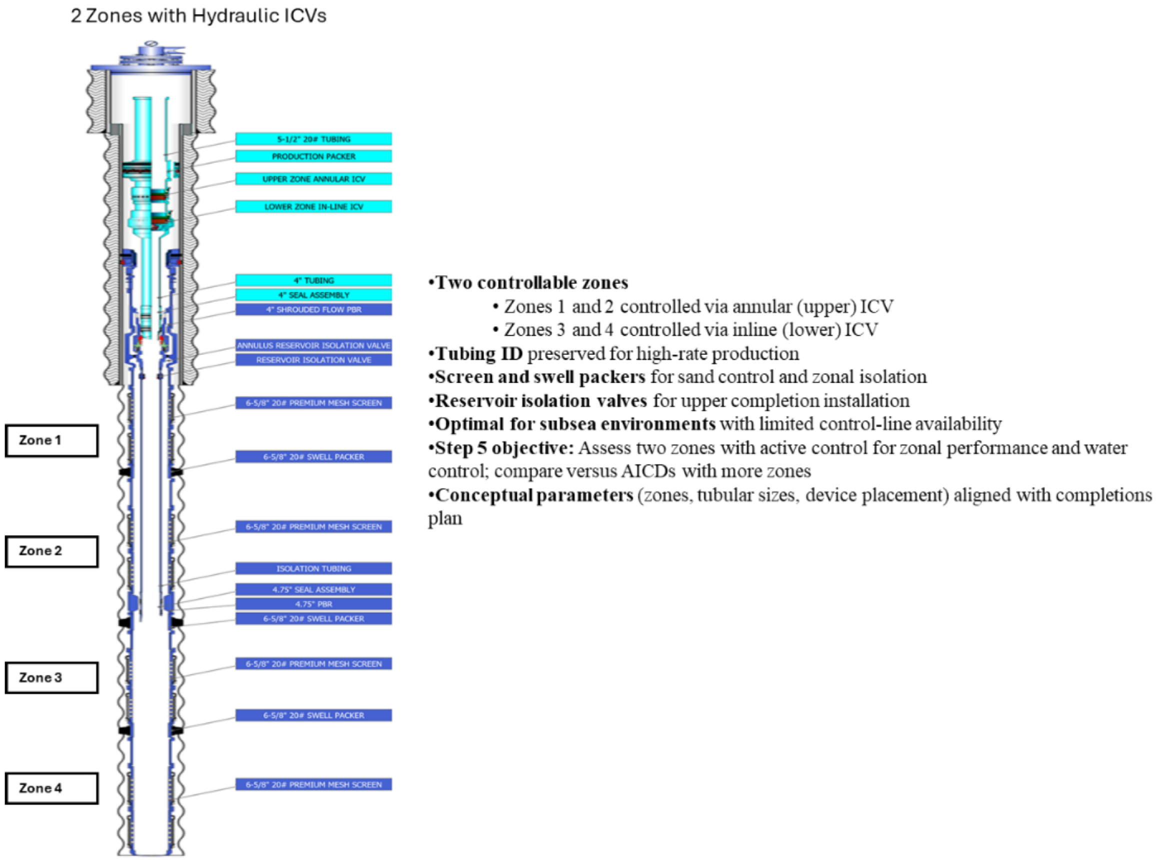


Figure 6—Sample conceptual design selected for validation.

Step 5: Validate via Modeling and Field Examples

Will the selected technologies deliver in real wells?. Unlike traditional workflows that model early and screen later, the proposed framework applies validation only to options that have already passed strategic and technical screening, thus focusing effort on viable configurations and avoiding wasted resources and late-stage redesigns.

At this stage, modeling confirms whether the short-listed technologies can meet the project's key performance-based (PB) requirements—such as drawdown balancing, breakthrough control, flow-assurance risk reduction, and adaptability—under realistic reservoir and operating conditions. These PB aspects, identified in Step 3, directly influence well performance, materials selection, and operational risk.

Depending on the project phase and data availability, validation can range from analytical approximations to high-resolution coupled simulations. Field analogs may be used alongside modeling to strengthen decision confidence, particularly where novel designs or operational constraints are involved.

Modeling Strategy and Integration of Constraints. The modeling strategy—analytical, semi-dynamic, well-level dynamic or reservoir-level dynamic— is chosen based on the project's maturity, data availability, and technology behavior.

Analytical (steady-state) modeling estimates zonal inflow distribution and heel-toe effects at a single point in time, under fixed reservoir and boundary conditions, making it useful for early-phase screening of passive systems such as ICDs when time or data is limited (Prakasa et al., 2019). However, it neglects dynamic effects, such as saturation changes or mobility evolution, and is not suitable for autonomous, adaptive, or active systems (Gurses and Vasper, 2013).

Semi-dynamic modeling updates steady-state inflow predictions using time-dependent pressure or saturation data from external reservoir models. These methods incorporate changes in pressure, mobility, and water cut over time but do not compute feedback from the completion system to the reservoir model. In the Alvheim Field, steady-state models were updated with time-dependent pressure data to compare ICD and AICD behavior. Laboratory tests, calibration, and tracer-based validation confirming performance. While this semi-dynamic method proved effective for design refinement and performance confirmation, the authors noted that such models inherit the uncertainty from input cases and cannot simulate redistribution or dynamic feedback (Langaas et al. 2019).

Dynamic well-level modeling uses multisegmented well (MSW) architecture, focuses on the near-wellbore region to simulate device-level pressure drops, zone interactions, and control logic. It captures time-dependent changes, such as evolving water cut, gas/oil ratio (GOR), and drawdown balance, making it effective for optimizing ICD nozzle sizing, AICD choking behavior, or ICV actuation strategies in early planning or brownfield optimization (Gurses and Vasper, 2013; Basak and Gurses, 2015). While it cannot capture full interwell interference or facility impacts, well-level modeling complements higher-resolution reservoir simulations.

For example, AICD response to changing water cut and GOR was calibrated using modeled wells informed by analog data and device-specific performance logic (Gurses et al., 2019). In another case, MSW-based simulation supported optimal electric ICV configurations in a 10,000-ft extended-reach well, guiding valve placement and inflow balance (Basak and Gurses, 2015).

Dynamic reservoir-level modeling (full field or sector) simulates inflow control technology performance over time while fully coupling reservoir behavior, completion logic, and, when relevant, surface system constraints. Unlike well-level models, this scale captures interwell interference and systemwide impacts from water- or gas-handling limits, compression schedules, and backpressure. Surface coupling can be implemented as full integration with the production system or via externally imposed constraints. This approach is applicable to all inflow control technology types and is especially valuable when evaluating multiwell developments or complex fieldwide interactions.

In Peru's field, sector modeling guided cyclonic AICD placement; post-job tracer diagnostics confirmed the modeled inflow behavior and showed a 57% reduction in water cut (Acencios et al., 2023). In a fractured carbonate reservoir, sector modeling with an advanced completion optimizer (ACO) compared ICDs, adaptive ICDs, and hydraulic/electrohydraulic ICVs over 20 years, showing that hybrid completions improved recovery, delayed breakthrough, and reduced opex (Amjad et al., 2020). Similarly, an ACO workflow optimized ICD and AICD placement in a heterogeneous carbonate with variable injector support, improving inflow balance and sweep efficiency (Elfeel et al., 2021).

Reservoir-level modeling also improves robustness by running multiple geological realizations, avoiding overconfidence from a single-deterministic model. The resulting probabilistic productivity ranges help asset decisions on ultimately well count and capex spend. In a carbonate oil-rim, cloud-based optimization of electric ICVs with continuous displacement chokes was tested across two geological realizations, including a high-permeability channel scenario. The optimization mitigated rapid water influx risk by dynamically adjusting drawdown, maintaining oil production under both cases (Gurses et al., 2025).

Qualitative Assessment of Operational Risks. Operational risks, such as erosion, plugging, and mechanical failure, should be assessed alongside performance. ICD and AICD erosion is driven by velocity, particle loading, and operating conditions, while solids accumulation can cause localized plugging (Chochua et al., 2018). ICVs add risks tied to actuator reliability, control line integrity, and seal performance, especially under frequent cycling. Hybrid and adaptive systems introduce additional failure modes, such as sleeve jamming or hydraulic leaks.

Risk evaluation can be done in stages. First, screen with analog performance data, design limits (e.g., velocity and sand-rate envelopes), and material qualification. Where predicted drawdown, solids, or novel geometries indicate elevated risk, perform targeted laboratory testing and use computational fluid dynamics (CFD), geomechanical, or flow-assurance modeling to quantify erosion, hydrate formation in both near-wellbore and surface systems, reservoir damage, or other degradation mechanisms. Integrating this risk assessment into Step 5 ensures that only designs with both technical merit and operational robustness advance to Step 6 for examining execution readiness (Fig. 7).

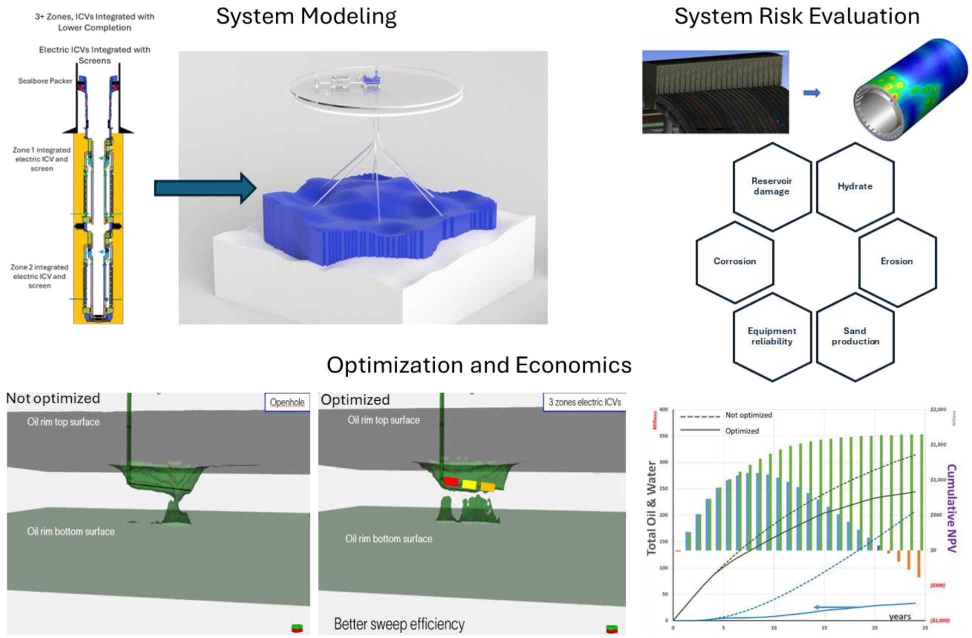


Figure 7—Integrated validation workflow in Step 5.

Modeling Device-Specific Behaviors. Each inflow control technology has unique modeling requirements, based on how it interacts with evolving subsurface and operational conditions.

ICDs are highly sensitive to reservoir heterogeneity. In Fig. 8, applied to a heterogeneous reservoir under uncertainty, a uniform ICD design achieved only moderate reductions in gas and water production and reduced oil recovery. By contrast, a customized configuration based on reliable geological description delivered a more effective drawdown balance and improved sweep efficiency over time. This underscores the importance of accurate reservoir knowledge for an effective ICD design strategy; over-choking or poor compartmental tuning can accelerate breakthrough and lower recovery.

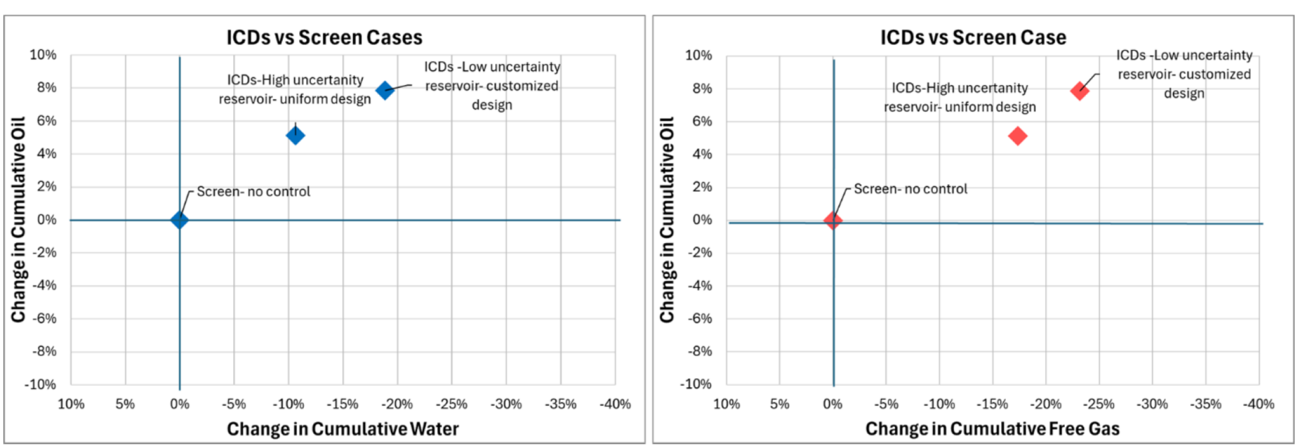


Figure 8—Impact of reservoir certainty on ICD design strategy and, consequently, oil recovery and production of unwanted phases.

AICDs number and placement also significantly influence inflow. As shown in Fig. 9, in this light-oil case the viscosity contrast to water was small, reducing phase selectivity and limiting incremental oil gain. Placement strategy, however, made a substantial difference: per-joint installation gave limited improvement, alternate-joint layouts improved inflow balance, and compartment-specific designs delivered the best recovery. This highlights the risk of relying on a single deterministic model, which may misrepresent device performance if actual reservoir conditions differ. Ensemble evaluation of device count and placement is therefore essential to capture sensitivity and deliver robust designs under uncertainty.

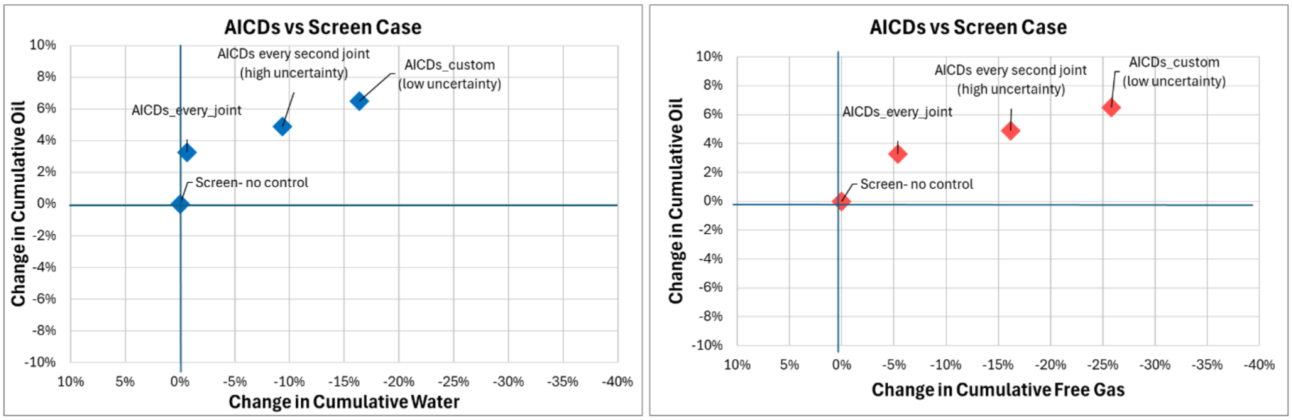


Figure 9—Impact of AICD design strategy on oil recovery and control of unwanted phases in a light-oil example.

ICVs, adaptive and hybrid systems require time-coupled modeling of control logic, especially in the presence of breakthrough, zonal interaction, or variable surface constraints. In one extended-reach application, Continuous-choke ICVs adapted to saturation changes and delayed water entry more effectively than their fixed-choke counterparts (Basak and Gurses, 2015). On/off ICDs with a shiftable internal sleeve can function as an adaptive system by shutoff for high-water-cut zones without replacing the completion (Amjad et al., 2020). Integrating surface water-handling limits into ICV actuation models further improved zone rebalancing and production continuity (Taksaudom et al., 2021). In a carbonate oil rim, dynamic modeling of electric ICVs with displacement chokes coupled with saturation tracking, surface capacity limits, and actuation logic guided control refinement and reduced water production by 74% (Gurses et al., 2025). For hybrid systems, modeling must account for how the relative contribution of each device type can change over time (e.g., an ICD handling inflow balancing while an ICV selectively manages breakthrough). This requires evaluating multiple scenarios with different reservoir conditions to confirm that both components will function synergistically when required.

Role of Empirical Validation and Cross-Disciplinary Collaboration. Analog field data complements modeling, especially for novel technologies such as mechanical AICDs or hybrid completions. Field trials have shown that AICDs can reduce gas or water production in complex reservoirs and that electric ICVs empower dynamic production optimization under variable conditions.

Effective validation requires close collaboration across subsurface, production, and completions teams, ensuring that models reflect realistic control logic and device behavior. Customization of existing workflows and access to scalable computing resources enable broad scenario testing, providing stakeholders with risk-informed, performance-based justification for inflow control technology selection.

Step 6: Finalize Completion

Are we ready to install, monitor, and adapt?

Based on screening and validation, the multidisciplinary team confirms the completion design or a small set of pilot options, aligned with reservoir uncertainties, production objectives, infrastructure limits, sandface design requirements, zonal control needs, and life-of-field goals. Integration with surface and subsea systems is verified, addressing control line availability, wellhead space, and chemical injection compatibility. The selected design(s) must be executable, monitorable, and adaptable to changing downhole conditions, with reliability and intervention strategies defined to ensure sustained performance over the life of the field.

Case Study: Field Development Strategy A

This case study involves a subsea greenfield in the North Sea, targeting a sandstone reservoir. Long horizontal producers are planned to recover light oil from a complex three-phase system (gas, oil, and water), with injected gas fully miscible with the reservoir oil. Pressure support is provided by gas injectors and an aquifer. The reservoir has significant permeability contrast, with channelized flow units creating a high risk of early gas or water breakthrough.

Early in development, subsurface data was limited, and uncertainties were high. Reservoir simulation was constrained to a single geological realization, restricting probabilistic analysis, but results were interpreted in the context of uncertainty to support screening and validation of completion strategies. This setting provided a rigorous test for the technology selection framework, especially in balancing recovery optimization and uncertainty mitigation.

Step 1: Project Drivers. The development strategy is guided by three primary objectives.

- **Maximize oil recovery** and sweep efficiency.
- **Prioritize gas control over water control** to remain within surface gas-handling capacity and minimize flaring in line with sustainability and regulatory targets.
- **Operate within strict subsea infrastructure constraints**, including limited control line capacity, power supply, and intervention access.

These drivers were reviewed and agreed with all relevant stakeholders, ensuring alignment across subsurface, facilities, and operations teams. They define the decision frame for subsequent screening, ensuring that inflow control options are assessed for their ability to manage gas breakthrough risk, optimize oil recovery, and fit the subsea system's limitations.

Step 2: Key Uncertainties and Critical Factors. Field planning is influenced by several subsurface and system-level uncertainties:

- timing and location of gas and water breakthrough.
- variability in fluid mobility and sweep efficiency, especially under miscible gas conditions.
- zonal heterogeneity and potential crossflow between zones, affecting conformance control and breakthrough progression.

In addition, several critical constraints amplify the impact of these uncertainties:

- maximum of four isolation zones, dictated by packer limitations.
- maximum of four control lines available because of subsea infrastructure restrictions.
- no downhole intervention capability in the subsea setting.
- constrained gas-handling capacity, requiring proactive gas control to maintain oil production targets.

These uncertainties and constraints were carried forward into Step 3 to guide the prioritization of screening aspects.

Step 3: Select Aspects for Screening. High-priority aspects included tolerance for reservoir uncertainty, adaptability to evolving mobility contrasts, zonal isolation capability, equipment reliability, and compatibility with subsea capacity. Gas control was prioritized over water control as explained above. These aspects formed the input to Step 4.

Table 5—Priority of each screening aspect for field development strategy A.

Technical Domains	Potential Key Screening Aspects	Screening Role	Strategy A: Max. Recovery
Reservoir engineering	1. Uncertainty in reservoir description	PB	High
	2. More flexible development	PB	High
	3. Number of controllable zones	PB	High
	4. Formation permeability	PB	Medium
	5. Well type (horizontal, multilateral)	C/EB	High
Completions engineering	6. Tubing size (ID)	C/EB	High
	7. Value of information (monitoring)	C/EB	High
	8. Zonal isolation (shutoff)	C/EB	High
	9. Equipment reliability	C/EB	High
Production engineering	10. Control of gas	PB	High
	11. Control of water	PB	Medium
	12. Production or injection rate variation	PB	High
	13. Max. production or injection rate	C/EB	High
	14. Well cleanup efficiency	PB	Medium
	15. Acidizing and scale treatment	PB	Medium
Economics & operations	16. Equipment cost	C/EB	Medium
	17. Installation complexity	C/EB	Medium
	18. Running-in-hole risk	C/EB	Medium
	19. Reservoir isolation during UC	C/EB	Medium
	20. Intervention requirement (design to eliminate)	C/EB	High
	21. Installation time	C/EB	Medium
	22. Plugging risk	C/EB	Medium
	23. Logistics constraints	C/EB	Medium
Infrastructure: Surface & subsurface	24. Subsea interfacing	C/EB	High
	25. Subsea capacity constraints	C/EB	High
	26. Flow assurance risks	PB	High
	27. Corrosion and erosion risks	PB	High

Step 4: Screening and Comparison. A broad set of inflow control options—including passive, autonomous, adaptive, active, and hybrid configurations—was evaluated against Strategy A's prioritized screening aspects (Table 6). Standard screens (no inflow control) were used as the baseline case to quantify the value of alternative strategies.

Table 6—Comparative evaluation of each inflow control system across the screening aspects critical to strategy A.

Discipline/Domain	Potential Key Screening Aspects	Strategy A: Max. Recovery	Screening Role	Standard Screens	Passive ICDs	Adaptive ICDs	Autonomous ICDs	2 Zones, Hydraulic ICVs *	2 Zones, Hydraulic ICVs *+ subzones, Passive ICDs	3+ Zones Electro-hydraulic ICVs*	3+ Zones Electric ICVs*
Reservoir engineering	1. Uncertainty in reservoir description	High	PB	🔴	🟡	🟡	🟡	🟡	🟢	🟢	🟢
	2. More flexible development	High	PB	🔴	🔴	🔴	🟡	🟡	🟢	🟢	🟢
	3. Number of controllable zones	High	PB	🔴	🟢	🟢	🟢	🟡	🟢	🟢	🟢
	5. Well type (horizontal, multilateral)	High	C/EB	🔴	🟢	🟢	🟢	🟢	🟢	🟢	🟢
Completions engineering	6. Tubing size (ID)	High	C/EB	🟢	🟢	🟢	🟢	🟢	🟢	🔴	🟢
	7. Value of information (monitoring)	High	C/EB	🔴	🔴	🔴	🔴	🟡	🟡	🟢	🟢
	8. Zonal isolation (shutoff)	High	C/EB	🔴	🔴	🟢	🔴	🟢	🟢	🟢	🟢
	9. Equipment reliability	High	C/EB	🟢	🟢	🟡	🟢	🟡	🟡	🟡	🟡
Production engineering	10. Control of gas	High	PB	🔴	🟡	🟢	🟢	🟢	🟢	🟢	🟢
	11. Control of water	Medium	PB	🔴	🟢	🟢	🟢	🟢	🟢	🟢	🟢
	12. Production or injection rate variation	High	PB	🔴	🟡	🟡	🟢	🟡	🟡	🟢	🟢
	13. Max production or injection rate	High	C/EB	🟡	🟢	🟢	🟢	🟢	🟢	🟡	🟢
Economics & operations	16. Equipment cost	Medium	C/EB	🟢	🟡	🟡	🟡	🔴	🔴	🔴	🔴
	20. Intervention requirement (design to eliminate)	High	C/EB	🔴	🔴	🟢	🔴	🟢	🟢	🟢	🟢
	21. Installation time	Medium	C/EB	🟢	🟢	🟢	🟢	🟡	🟡	🟡	🟡
	23. Logistics constraints	Medium	C/EB	🟢	🟢	🟢	🟢	🟡	🟡	🟡	🟡
Infrastructure: Surface & Subsurface	24. Subsea interfacing	High	C/EB	🟢	🟢	🟢	🟢	🟢	🟢	🟢	🟢
	25. Subsea capacity constraints	High	C/EB	🟢	🟡	🟢	🟡	🟢	🟢	🟢	🟢
	26. Flow assurance risks	High	PB	🔴	🔴	🟡	🟡	🟡	🟡	🟡	🟡
	27. Corrosion and erosion risks	High	PB	🟡	🟡	🟡	🟡	🟡	🟡	🟡	🟡

****** Hydraulic and electrohydraulic ICVs = discrete 8-position control; Electric ICVs = continuous control***

All technologies, including those expected to underperform, are retained during initial screening to ensure comparative evaluation and capture trade-offs relevant to reservoir complexity, uncertainty, and infrastructure constraints.

Summary of Screening Results

- **Standard screens** provide no inflow control and are prone to early heel breakthrough in high-permeability channels and heterogeneous reservoirs. **Eliminated.**
- **ICDs** rely on a fixed design that lacks adaptability to evolving mobility contrasts or miscibility effects, and has no zonal shutoff under high uncertainty and channeling risk. **Eliminated.**
- **Adaptive ICDs** shutoff capability is beneficial, but operational complexity and Strategy A's lack of tolerance for interventions make these systems impractical. **Eliminated.**
- **Mechanical AICDs** miscible gas reduces viscosity contrast, making phase selectivity uncertain under multiphase flow and heterogeneity. Placement sensitivity further limits their suitability. **Eliminated.**
- **Hydraulic ICVs (2 zones)** offer proven active zonal control but are constrained by control line availability, limiting deployment to two zones. Their use is viable only in naturally compartmentalized sections, such as channelized or shale-bounded zones. **Conditionally retained** for niche applications.
- **Hybrids (ICVs + ICDs)** combine zonal control with intrazone balancing, reducing control line demand. They lack sub-zone selectivity, increasing in-zone breakthrough risk, and require careful placement evaluation. **Retained.**
- **Electrohydraulic ICVs (3+ zones)** enable multiplexed actuation with fewer control lines, but the inner-string architecture reduces tubing ID, limiting capacity in high-rate wells. **Eliminated.**
- **Electric ICVs (3+ zones)** provide scalable, real-time zonal control without hydraulic line constraints, maintain fullbore ID for high-rate production, and align with subsea infrastructure limits. They are well-suited for high-uncertainty, high-flow, heterogeneous reservoirs; higher cost is to be justified in Step 5. **Retained.**

Conceptual Designs Created for Validation

Four configurations progressed to Step 5 for evaluation:

- **ICDs (baseline)** as the reference case to quantify the value of alternative inflow control strategies under equivalent conditions
- **Two-zone hydraulic ICV systems**
- **Hybrid configurations (ICVs + ICDs)**
- **Fully electric ICVs (3+ zones),**

These configurations with conceptual completion designs (Fig. 10) formed the input for modeling workflows, where their performance was assessed under representative subsurface and operational scenarios.

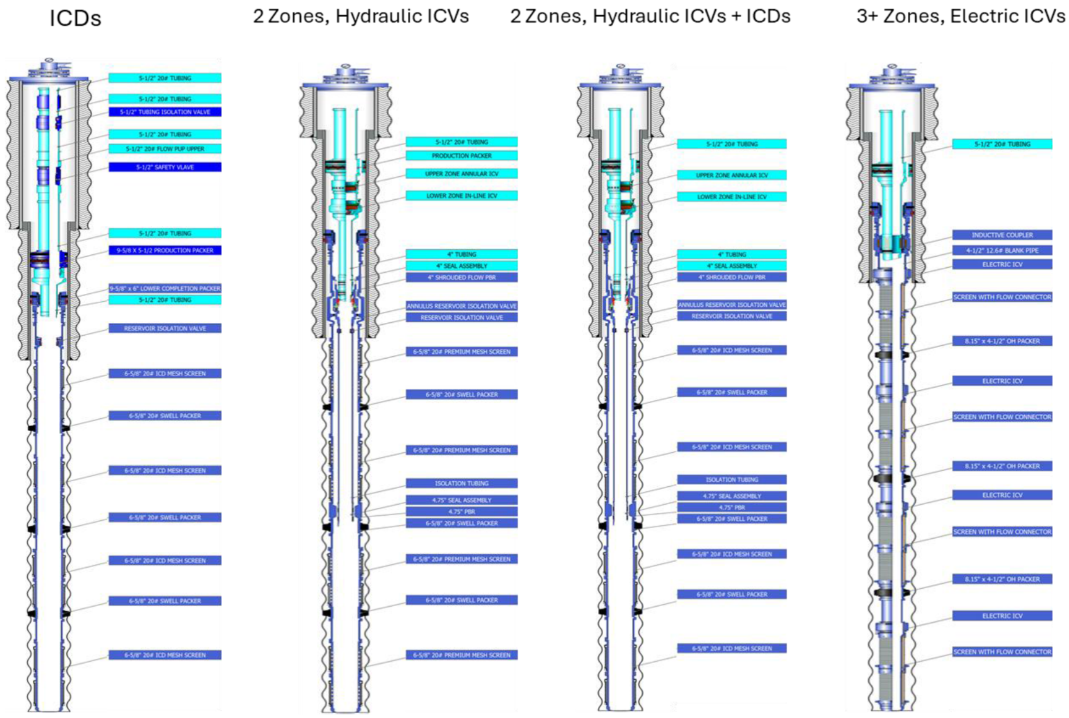


Figure 10—Conceptual completion designs for field development strategy A.

Step 5: Evaluation via Modeling. Reservoir-level dynamic modeling was performed on two ~2000-m horizontal oil producers and a gas injector completed with standard screens. The four short-listed technologies from Step 4 were tested under identical reservoir and operating conditions to quantify incremental oil recovery and changes in gas and water production.

Dynamic modeling confirmed that all the other short-listed configurations would outperform the ICD base case in both oil recovery and reduction of unwanted fluids, but to varying extents. Fig. 11 shows that the 4-zone electric ICVs delivered the highest oil uplift (7%) and the largest reduction in free gas (–34%) and water (–15%), leading to the highest net present value (NPV). The hybrid 2-zone ICV + ICD design ranked second, offering strong gas and water control with slightly lower oil gain. Two-zone hydraulic ICVs provided moderate improvements but limited zonal control capability. Overall, the 4-zone electric ICVs best balance reservoir performance, infrastructure compatibility, and economic return.

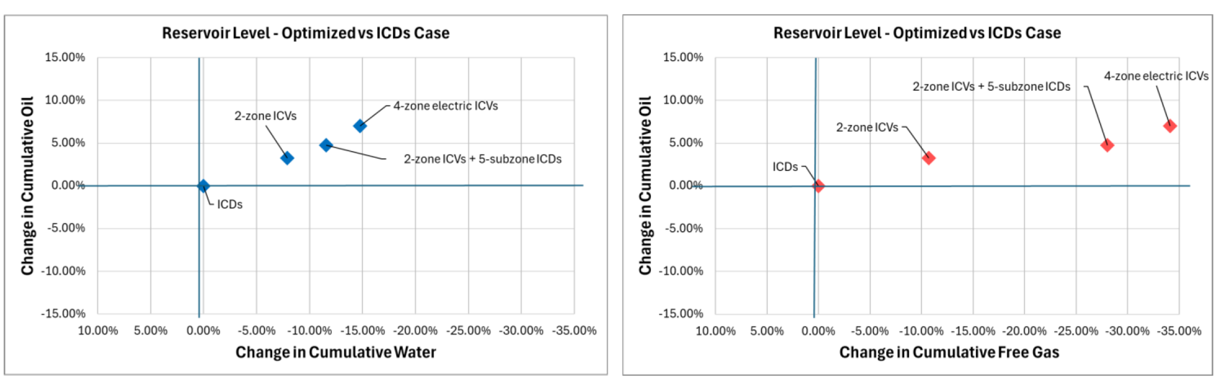


Figure 11—Modeled change in cumulative oil recovery versus reduction in free gas (left) and water production (right) relative to the ICD base case for the four short-listed inflow control completion designs over a 10-year period.

Step 6: Finalize Completions. Given the high level of reservoir and operational uncertainty—especially regarding breakthrough timing, fluid mobility variation, and heterogeneity—dynamic modeling with a single geological realization was used as an indicative, rather than definitive, predictor of inflow control performance. While the 4-zone electric ICVs achieved the highest modeled oil gain, greatest reduction in free gas and water, and highest NPV, the single-realization results do not fully capture the full range of possible reservoir behaviors.

To derisk the completion strategy, two configurations were chosen for field trials in early wells:

- **4-zone electric ICVs,** providing high-resolution zonal control, fullbore design, no hydraulic line constraints, and maximum adaptability if zonal inflow patterns differ from forecasts
- **hybrid 2-zone hydraulic ICVs + ICDs,** combining targeted zonal control with intrazone flow balancing, reducing control line demand and offering mechanical simplicity, which may prove advantageous if electric ICV deployment or control integration encounters constraints.

Performance of both configurations will be monitored in the initial campaign to validate subsurface models, assess control system responsiveness under real conditions, and refine the longer-term completion selection for full-field rollout.

Fig. 12 shows representative completion designs for the selected configurations. Each design incorporates trade-offs between adaptability, mechanical complexity, and surface control constraints.

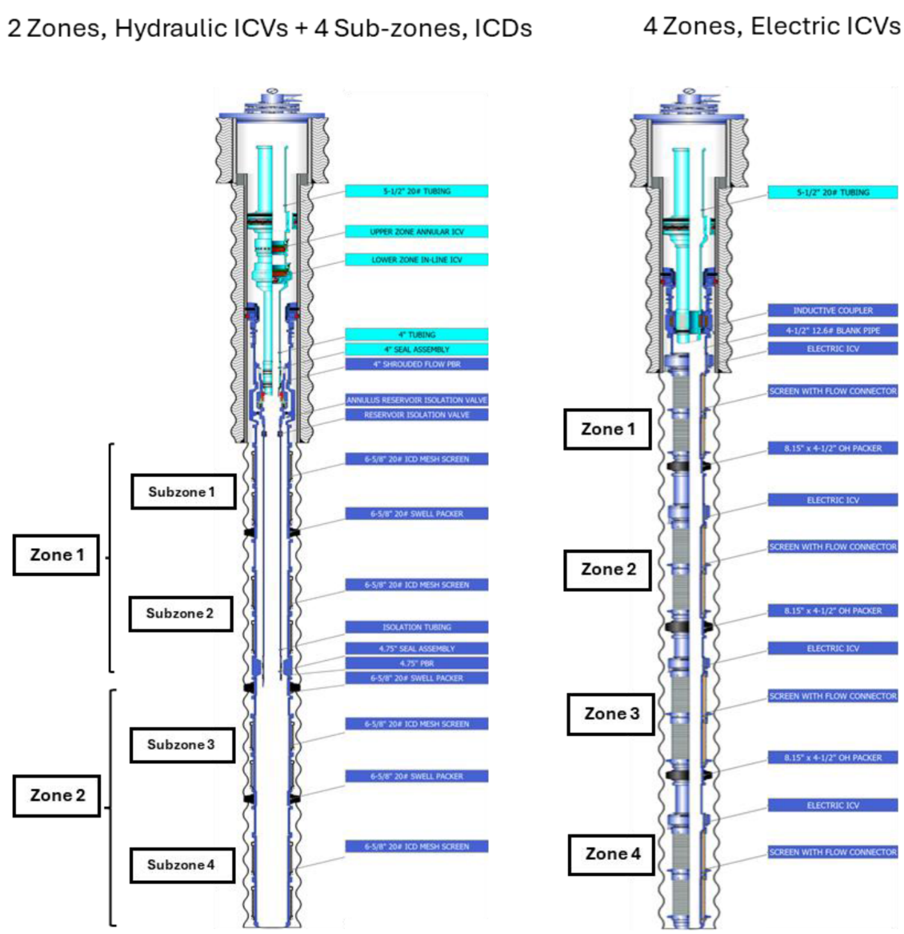


Figure 12—Sample completion designs incorporating the selected inflow control technologies.

Discussion

Growing subsurface complexity, variable surface constraints, and the need for optimized production outcomes demand more integrated, scalable workflows for inflow control technology selection. The six-step framework introduced in this paper enables structured, multidisciplinary evaluation, and its impact can be further enhanced by embedding digital tools, data-driven analytics, and AI-assisted decision support.

In Step 3, prioritization of screening aspects benefits from a digital tool that formalizes the evaluation of PB and C/EB. It could allow multidisciplinary teams to map project objectives and operational limits into structured decision matrices. AI-driven algorithms that include rule-based engines and multicriteria ranking models can accelerate convergence by highlighting viable technology pathways and eliminating unfit options early.

Although reservoir uncertainty is recognized as a PB aspect with significant upside in recovery or unwanted fluid control, technologies that could capture this value are often eliminated early in concept development. Offset analogs, lookback studies, and Step 5 modeling are therefore critical to improve confidence and support informed investment decisions.

For data-rich or high-uncertainty developments, such as stacked reservoirs or multiwell fields, machine learning (ML) models trained on analogous projects or synthetic simulations can aid in anticipating inflow behavior, identifying outliers, and clustering similar conditions. This probabilistic insight enhances shortlisting accuracy and supports scenario-driven sensitivity analysis. While ML is not a substitute for engineering judgment, it complements traditional screening methods and improves consistency at scale.

Step 5 is strengthened by integrated modeling platforms that couple inflow control logic with dynamic reservoir and surface system simulations. Ensemble workflows allow teams to test technology behavior across multiple geological realizations and operational constraints, validating whether shortlisted systems maintain performance under uncertainty. This is especially critical when evaluating high-cost or intervention-limited designs.

Integration needs vary by setting. Subsea or offshore environments require early coordination across domains because of infrastructure limitations and access constraints. In contrast, onshore fields may enable more adaptive strategies. Regardless of setting, embedding digital tools enables more predictive, transparent, fit-for-purpose inflow control decisions and supports execution-ready design.

While monitoring systems are not the primary focus of this paper, they are a critical enabler of effective inflow control, especially in hybrid designs, staged field developments, or environments with limited access. Technologies such as distributed fiber-optic sensing, sandface-deployed pressure gauges, and chemical tracers can be integrated with passive, autonomous, adaptive, or active completions to enhance zonal diagnostics and flow verification. In subsea or intervention-constrained settings, such surveillance enables real-time or near-real-time insights into production and injection distribution, supporting dynamic reservoir characterization and control logic refinement over time. As digital workflows mature, integrating inflow control selection with monitoring design will become increasingly important to maximize system value and inform long-term field performance optimization.

Conclusions

This paper introduces a structured six-step framework that supports early, strategy-aligned inflow control technology selection by systematically integrating subsurface complexity, well completion design, and surface system constraints. It moves inflow control from isolated component design toward a coordinated decision process shared across reservoir, completions, production, and facilities teams.

The framework distinguishes PB screening aspects—such as zonal isolation capability, response to production/injection rate variation, and sensitivity to reservoir description uncertainty—from C/EB filters, such as intervention requirements, subsea infrastructure limitations, and installation complexity. This screening logic helps eliminate unfit technologies at an early stage, focusing engineering efforts on options that align with project objectives such as adaptability to changing reservoir conditions, drawdown management, and control of unwanted fluid.

Comparative evaluation of passive, autonomous, adaptive, and active inflow control technologies illustrates that no single solution fits all contexts. Trade-offs must be balanced based on uncertainty tolerance, both upside and down, control resolution, intervention strategy, and surface integration requirements. By aligning inflow control decisions with the full production system, the framework enhances completion design quality, reduces late-stage redesign risk, and delivers execution-ready solutions that optimize performance across the life of the field, supporting the industry's broader goals of integrated, high-impact field development and sustained operational efficiency.

Acknowledgments

The author thanks SLB for permission to publish this work.

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